University of Hull
Department of Mathematics

Dr. Silvio Fanzon
S.Fanzon@hull.ac.uk

Differential Geometry

Homework 3

Please upload your solutions on Canvas by 8th November 2023 at 14:00 hours

The total number of marks awarded is 100

Q1. Describe all curves $\boldsymbol{\gamma}\colon \mathbb{R} \to \mathbb{R}^3$ which have constant curvature κ and torsion τ, with

$$\kappa = \frac{1}{2}\,, \quad \tau = \frac{1}{2}\,.$$

Hint: This problem does not require any calculation. Think about the results and examples we have seen in Chapter 3 of the Lecture Notes.

[10 marks]

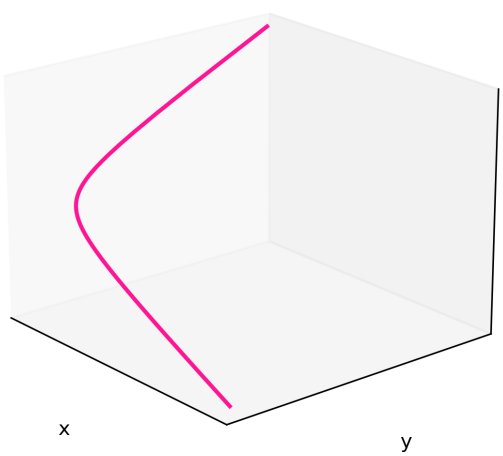

Figure 1: Curve $\boldsymbol{\gamma}$ from Q2

Q2. Define the curve $\boldsymbol{\gamma}\colon (0,\infty) \to \mathbb{R}^3$ by

$$\boldsymbol{\gamma}(t) := \left(t, \frac{1+t}{t}, \frac{1-t^2}{t}\right)\,.$$

(i) Prove that $\boldsymbol{\gamma}$ is regular.

[5 marks]

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(ii) Let κ denote the curvature of $\boldsymbol{\gamma}$. Prove that $\kappa(t) \neq 0$ for all $t > 0$.
Hint: You do not need to carry out the full calculation for κ.

[5 marks]

(iii) Let τ denote the torsion of $\boldsymbol{\gamma}$. Prove that $\tau(t) = 0$ for all $t > 0$.
Hint: You do not need to carry out the full calculation for τ.

[5 marks]

(iv) Prove that $\boldsymbol{\gamma}$ is contained in a plane.
Hint: You can use one of the results in Chapter 3 of the Lecture Notes.

[5 marks]

Q3. Consider the curve $\boldsymbol{\gamma}\colon \mathbb{R} \to \mathbb{R}^3$ defined by

$$\boldsymbol{\gamma}(t) = (1 + \cos(t), 1 - \sin(t), \cos(2t)) \,.$$

(i) Prove that $\boldsymbol{\gamma}$ is regular.

[5 marks]

(ii) Compute the curvature of $\boldsymbol{\gamma}$.

[10 marks]

(iii) Compute the torsion of $\boldsymbol{\gamma}$.

[10 marks]

Hint: Use that $\cos(2t) = 2\cos^2(t) - 1$.

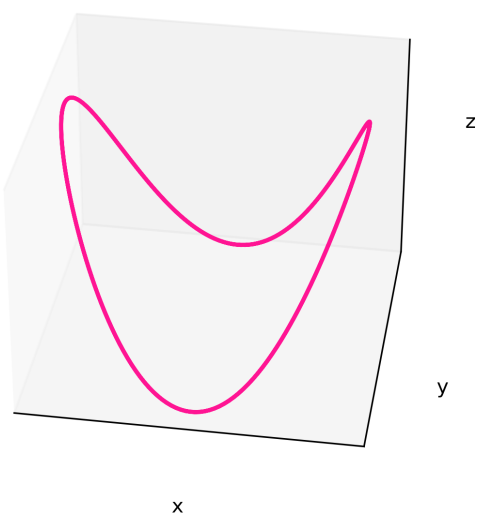

Figure 2: Curve $\boldsymbol{\gamma}$ from Q3

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Q4. Let $(X, \mathcal{T})$ be a topological space and $x_0 \in X$. Define the sequence

$$x_n := x_0 \,, \quad \forall\, n \in \mathbb{N} \,.$$

Prove that $x_n \to x_0$.

[5 marks]

Q5. Let (X, d) be a metric space and $\mathcal{T}_d$ the topology induced by d. Then:

(i) Prove that, for all $x \in X$, $r > 0$, we have $B_r(x) \in \mathcal{T}_d$.

[10 marks]

(ii) Prove that $U \in \mathcal{T}_d$ if and only if

$$U = \bigcup_{i \in I} B_{r_i}(x_i) \,,$$

with I suitable family of indices and $x_i \in U$, $r_i > 0$.

[10 marks]

Q6.

(i) Let (X, d) be a metric space. Denote by $\mathcal{T}_d$ the topology induced by d and let $A \subseteq X$. Prove that

$$\operatorname{Int} A = \{x \in A : \exists\, r > 0 \text{ s.t. } B_r(x) \subseteq A\} \,.$$

[10 marks]

(ii) Let $X = \mathbb{R}$ with d the Euclidean metric. Let $A = [0, 1)$. Prove that

$$\operatorname{Int} A = (0, 1) \,, \quad \overline{A} = [0, 1] \,,$$

You may use any result from the Lecture Notes, as well as Q6 Point (i).

[10 marks]